

Lecture 14

The quantum inverse scattering method, and its classical limit

Objectives. Course map: G2 and G5, P2 quantised. [Lecture 10](#) solved the Heisenberg chain by a guess and left the question of why it is solvable. This lecture answers with [Lecture 9](#)'s three moves, quantised. The chain's Lax operator satisfies an RLL relation with Yang's R -matrix of [Lecture 13](#), the monodromy satisfies RTT, and the transfer matrices commute at all values of the spectral parameter. Then the lecture restores $\hbar$ in the quantum Toda chain and recovers [Lecture 9](#)'s r -matrix as $\hbar \rightarrow 0$, sign for sign.

14.1 The question

[Lecture 9](#) made the Toda ring's integrals commute with one identity for one site; [Lecture 10](#) solved the Heisenberg chain by a guess. So the question is

why do the Heisenberg chain's charges commute, and what classical structure is this the quantisation of?

The answer: [Lecture 9](#)'s structure, with commutators for brackets and Yang's R for r ([Theorems 14.1 to 14.3](#)); and the classical limit of the quantum Toda chain proves that the dictionary is right ([Corollary 14.6](#)). The worked example is the Heisenberg chain's Lax operator.

14.2 From brackets to commutators

[Lecture 9](#)'s site matrices obey the quadratic bracket (9.9); it composes along an ultralocal chain ([Theorem 9.7](#)), and its trace makes the transfer matrices Poisson-commute ([Corollary 9.8](#)). Quantisation makes the entries of L operators and brackets commutators, $\{\cdot, \cdot\} \rightarrow [\cdot, \cdot] / i\hbar$, and the legs of [Definition 9.2](#) survive. The quantum relation for the monodromy is the RTT relation

$$R_{12}(\lambda - \mu) T_2(\mu) T_1(\lambda) = T_1(\lambda) T_2(\mu) R_{12}(\lambda - \mu), \quad R = \mathbb{1} + i\hbar r + O(\hbar^2). \quad (14.1)$$

At order $\hbar$ it returns (9.9); with the other order of the monodromies it would return $r \rightarrow -r$, so the order is forced (§ 14.6). Other books use the other order, each with its own R and Lax operator: convert all three together, never one alone. Until § 14.6, $\hbar = 1$, and the auxiliary copies of $\mathbb{C}^2$ are a and b , apart from the sites $n = 1, \dots, N$.

14.3 The Lax operator of the Heisenberg chain

The quantum space is the Heisenberg chain of [Lecture 10](#), $\mathcal{H} = (\mathbb{C}^2)^{\otimes N}$, one spin $\frac{1}{2}$ per site. The auxiliary space a is one more $\mathbb{C}^2$. The *Lax operator* at site n is

$$L_{an}(\lambda) = \lambda + iP_{an} = \left(\lambda + \frac{i}{2} + iS_n^z \quad iS_n^- \right. \\ \left. iS_n^+ \quad \lambda + \frac{i}{2} - iS_n^z \right)_a, \quad (14.2)$$

with P_{an} the permutation of the auxiliary space with site n ([Definition 9.4](#)), $\mathbf{S} = \boldsymbol{\sigma}/2$ and $S^\pm = S^x \pm iS^y$. The middle form uses $P = \frac{1}{2}(\mathbb{1} + \boldsymbol{\sigma}_a \cdot \boldsymbol{\sigma}_n)$, the identity behind the Hamiltonian (10.2); the matrix is in the auxiliary space, with entries operators on site n . Like the Toda site matrix it is linear in the spectral parameter and acts on one site.

[Figure 14.1](#) draws it. Each space is a line: the auxiliary space horizontal, the site vertical. An operator on both is a vertex where they cross, and (14.2) says what happens there: with weight λ the two states go straight through, and with weight i they swap lines, the P of [Definition 9.4](#).

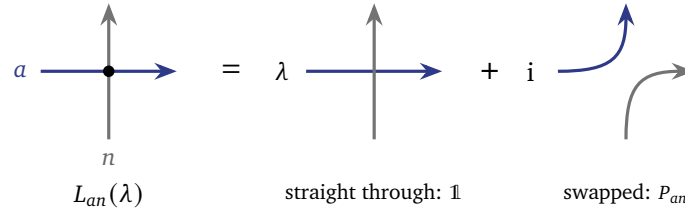


Figure 14.1: The Lax operator as a vertex. The auxiliary line (indigo) crosses the line of site n (grey). With weight λ both states go straight on; with weight i the auxiliary state leaves on the site's line and the site's state on the auxiliary line. (Where lines run side by side, a swap is drawn as a crossing; for lines that meet at right angles the crossing is the identity and the swap is two turns.)

Two auxiliary spaces a, b and one site n make three copies of $\mathbb{C}^2$, and on them the only operators needed are the three transpositions P_{ab}, P_{an}, P_{bn} . Their products are permutations of three objects, and a product of two different transpositions is a 3-cycle, of which there are two:

$$C = P_{ab}P_{an} = P_{an}P_{bn} = P_{bn}P_{ab}, \quad C^{-1} = P_{an}P_{ab} = P_{ab}P_{bn} = P_{bn}P_{an}. \quad (14.3)$$

(Follow a vector $x \otimes y \otimes z$ through each product; this is the computation of [Proposition 13.4](#).)

Theorem 14.1 (The RLL relation). With $R(\lambda) = \lambda - iP$,

$$R_{ab}(\lambda - \mu) L_{bn}(\mu) L_{an}(\lambda) = L_{an}(\lambda) L_{bn}(\mu) R_{ab}(\lambda - \mu) \quad (14.4)$$

for all λ, μ . With $\lambda + iP$ in place of R it fails.

Proof. Take $R = (\lambda - \mu) + siP_{ab}$ with a sign $s = \pm 1$ and compute the difference Δ of the two sides. It has two parts. The first is $(\lambda - \mu)(L_{bn}L_{an} - L_{an}L_{bn})$, and in $L_{bn}L_{an} - L_{an}L_{bn}$ everything cancels except the products of two P 's: $P_{an}P_{bn} - P_{bn}P_{an} = C - C^{-1}$. The second is

$$si(P_{ab}L_{bn}L_{an} - L_{an}L_{bn}P_{ab}) = si\left(i\mu(P_{ab}P_{an} - P_{an}P_{ab}) + i\lambda(P_{ab}P_{bn} - P_{bn}P_{ab}) - (P_{ab}P_{bn}P_{an} - P_{an}P_{bn}P_{ab})\right),$$

the terms $\lambda\mu P_{ab}$ having cancelled. By (14.3) the first bracket is $C - C^{-1}$, the second $C^{-1} - C$, and the triple products are both P_{bn} , so they cancel. The second part is therefore $s(\lambda - \mu)(C - C^{-1})$, and

$$\Delta = (1 + s)(\lambda - \mu)(C - C^{-1}).$$

This vanishes for $s = -1$ and is not zero for $s = +1$, since $C \neq C^{-1}$. ■

The whole relation is a statement about the permutations of three objects (the check script confirms the residual symbolically).

14.4 From one site to the chain

The monodromy matrix

The *monodromy matrix* of the chain is the ordered product of the Lax operators along it,

$$T_a(\lambda) = L_{aN}(\lambda) \cdots L_{a2}(\lambda) L_{a1}(\lambda) = \begin{pmatrix} A(\lambda) & B(\lambda) \\ C(\lambda) & D(\lambda) \end{pmatrix}_a, \quad (14.5)$$

an operator on $\mathbb{C}_a^2 \otimes \mathcal{H}$: in the auxiliary space a 2×2 matrix whose entries A, B, C, D are operators on the whole chain. As a picture it is a row: one auxiliary line crossing every site in turn (Figure 14.2), Figure 9.1 with operators at the vertices. Operators on different sites commute, so L_{an} commutes with L_{bm} for $n \neq m$: they act on disjoint sets of spaces. That is the quantum form of ultralocality, and it is all the next proof uses.

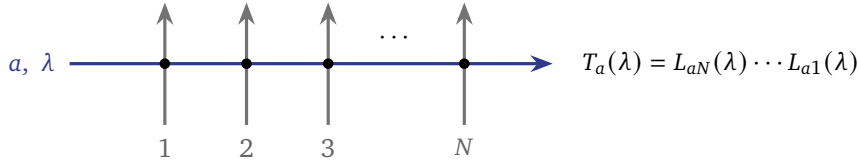


Figure 14.2: The monodromy matrix as a row. The auxiliary line meets site 1 first, so L_{a1} is the rightmost factor. The two free ends of the auxiliary line are the two indices of the 2×2 matrix (14.5); the ends of the site lines are the quantum space.

Theorem 14.2 (RLL implies RTT). *The monodromy (14.5) satisfies*

$$R_{ab}(\lambda - \mu) T_b(\mu) T_a(\lambda) = T_a(\lambda) T_b(\mu) R_{ab}(\lambda - \mu). \quad (14.6)$$

The same holds for any chain whose Lax operators obey (14.4) with a common R and act on different sites.

Proof. In $T_b T_a = L_{bN} \cdots L_{b1} L_{aN} \cdots L_{a1}$, move each L_{an} to the left past $L_{b,n-1}, \dots, L_{b1}$, with which it commutes. The factors pair up by site:

$$T_b(\mu) T_a(\lambda) = (L_{bN} L_{aN}) (L_{b,N-1} L_{a,N-1}) \cdots (L_{b1} L_{a1}).$$

Now R_{ab} enters on the left. By (14.4) at site N , $R_{ab} L_{bN} L_{aN} = L_{aN} L_{bN} R_{ab}$: it has passed the first pair and reversed it. The same step at each site in turn carries R_{ab} to the right end, leaving $(L_{aN} L_{bN}) \cdots (L_{a1} L_{b1}) R_{ab}$, and undoing the pairing gives $T_a T_b R_{ab}$. ■

This is the *train argument* (Figure 14.3): the crossing of the two auxiliary lines is pushed through the row one site at a time, as the bracket was pushed through the product by Leibniz's rule in Theorem 9.7.

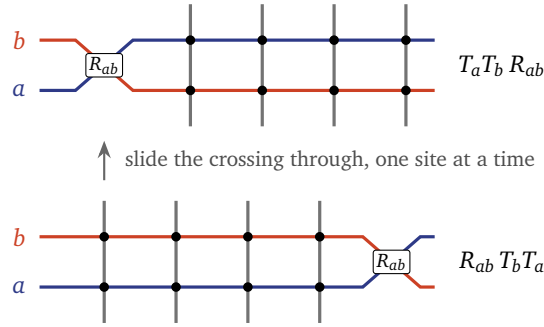


Figure 14.3: The train argument. Operators act left to right along the auxiliary lines, so the leftmost thing drawn is the rightmost factor. Bottom: the rows, then the crossing R_{ab} . By (14.4) the crossing slides through site N , then $N - 1$, $\dots$, until it has passed the row (top): (14.6).

The Lax operator is an R -matrix, and RLL is the Yang–Baxter equation

Compare (14.2) with R :

$$L_{an}(\lambda) = \lambda + iP_{an} = -R_{an}(-\lambda). \quad (14.7)$$

The Lax operator is the R -matrix itself, with the site as its second space. It is *not* $R_{an}(\lambda)$: the two differ by $\lambda \rightarrow -\lambda$. Substitute (14.7) into (14.4); the two minus signs cancel,

$$R_{ab}(\lambda - \mu) R_{bn}(-\mu) R_{an}(-\lambda) = R_{an}(-\lambda) R_{bn}(-\mu) R_{ab}(\lambda - \mu).$$

Rename $b \rightarrow 1$, $a \rightarrow 2$, $n \rightarrow 3$, and put $u = -\mu$, $v = -\lambda$, so that $\lambda - \mu = u - v$. Since $P_{12} = P_{21}$ this is

$$R_{12}(u - v) R_{13}(u) R_{23}(v) = R_{23}(v) R_{13}(u) R_{12}(u - v), \quad (14.8)$$

the *Yang–Baxter equation* in its standard form. With $u = u_{13}$, $v = u_{23}$ and $u - v = u_{12}$ it is the equation (13.6) that [Theorem 13.3](#) derived from three-body consistency, for the matrix $X_B(u) = R(u)/(u + i)$ of the delta bosons (13.3). So [Theorem 14.1](#) and [Proposition 13.4](#) are one fact, read twice. In [Lecture 13](#) the three lines were three particles and R a two-body collision; in the Heisenberg chain one of the three lines is a site. The fermions of [Lecture 13](#) scatter with $X_F = -L(u)/(u + i)$, the Lax operator itself.

[Figure 14.5](#) shows the equation as a picture. It is [Figure 13.1](#), and [Figure 14.4](#) is its special case with one line vertical.

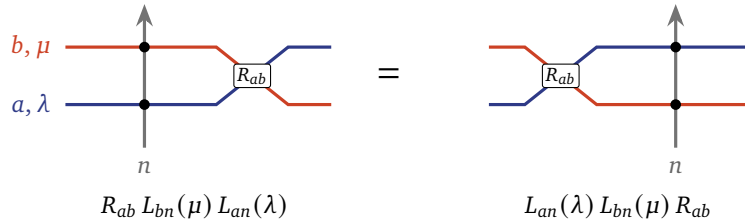


Figure 14.4: The *RLL* relation (14.4). Left: the site line n meets a (lower) and then b , and afterwards the auxiliary lines cross. Right: they cross first, and n meets b and then a . The crossing of a and b may be slid through the site line.

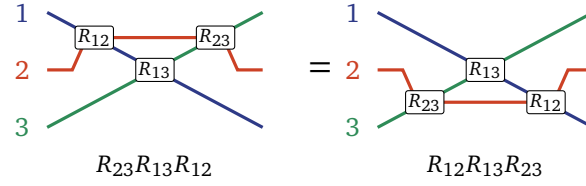


Figure 14.5: The Yang–Baxter equation (14.8) as a slide: line 2 passes either above or below the crossing of lines 1 and 3. Lines carry spectral parameters and a crossing of lines i, j carries R_{ij} at the difference of their parameters, which is why $u_{13} = u_{12} + u_{23}$. Reading left to right, the first crossing is the rightmost factor. Make line 3 a site and this is Figure 14.4.

14.5 Commuting transfer matrices

The *transfer matrix* is the trace of the monodromy over the auxiliary space,

$$t(\lambda) = \text{tr}_a T_a(\lambda) = A(\lambda) + D(\lambda), \quad (14.9)$$

an operator on the chain alone: in the picture the auxiliary line is closed into a loop (Figure 14.6). It is a polynomial of degree N in λ with operator coefficients, and leading term $2\lambda^N$.

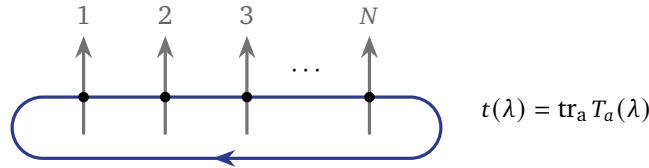


Figure 14.6: The transfer matrix: the auxiliary line of Figure 14.2, closed into a loop by the trace. The loop has no ends, so it acts on the sites alone; this is the loop of Figure 9.1 with operators at the vertices.

Theorem 14.3 (Commuting transfer matrices). $[t(\lambda), t(\mu)] = 0$ for all $\lambda, \mu \in \mathbb{C}$.

Proof. $R(u)R(-u) = (u - iP)(-u - iP) = -(u^2 + 1)$, so $R(u)$ is invertible for $u \neq \pm i$. For such $\lambda - \mu$, multiply (14.6) on the left by R_{ab}^{-1} : $T_b(\mu)T_a(\lambda) = R_{ab}^{-1}T_a(\lambda)T_b(\mu)R_{ab}$. Take the trace over both auxiliary spaces. On the left, $\text{tr}_{ab} T_b T_a = t(\mu)t(\lambda)$, because T_a and T_b have their auxiliary legs in different spaces. On the right, R_{ab} acts on the auxiliary spaces only, as a numerical matrix, and the partial trace is cyclic for such a factor; so $\text{tr}_{ab} (R_{ab}^{-1}T_a T_b R_{ab}) = \text{tr}_{ab} (T_a T_b) = t(\lambda)t(\mu)$. Both sides are polynomials in λ, μ , so the identity extends to $\lambda - \mu = \pm i$. ■

This is Corollary 9.8: there the trace of a commutator vanished, here the trace of a conjugate is the trace (Figure 14.7). The coefficients of $t(\lambda)$ commute with one another. Whether the family contains a Hamiltonian worth having, and whether its members are local, is the business of Lecture 15.

What RTT gives, and what it does not

The proof used (14.6) and the invertibility of R , and not the Yang–Baxter equation. The equation does two other jobs. It *supplies Lax operators*: by (14.7) the rational R , read with

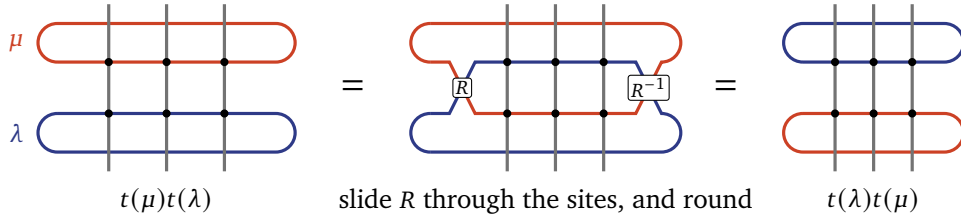


Figure 14.7: Why transfer matrices commute. Insert $RR^{-1} = \mathbb{1}$ between the loops; R slides through every site (Figure 14.3), exchanging the rows, and meets R^{-1} round the back. The loops have passed through each other.

a site as its second space, is a Lax operator for its own RLL relation, which is why the Heisenberg chain has one. And it is *sufficient, not forced*, for the consistency of the RTT relations taken as the definition of an algebra (the readings at the end). Finally, *integrability does not require RTT* : the free-fermion charges of the Ising chain (Lecture 18) are constructed directly.

14.6 The classical limit, verified: the quantum Toda chain

The Heisenberg chain has no classical limit worth the name — a spin $\frac{1}{2}$ has no small parameter. To test the correspondence of § 14.2, quantise Lecture 9's chain. Restore $\hbar$. The periodic Toda chain has canonical pairs with $[q_m, p_n] = i\hbar \delta_{mn}$, represented by $p_n = -i\hbar \partial/\partial q_n$ on functions of q . Keep the site matrix of (9.2) as it stands,

$$L_n(\lambda) = \begin{pmatrix} \lambda - p_n & e^{-q_n} \\ -e^{q_n} & 0 \end{pmatrix}, \quad (14.10)$$

now with operator entries. No ordering question arises: each entry involves q_n or p_n , never both.

Theorem 14.4 (Quantum Toda: RLL). *The site matrix (14.10) satisfies*

$$R_{ab}(\lambda - \mu) L_b(\mu) L_a(\lambda) = L_a(\lambda) L_b(\mu) R_{ab}(\lambda - \mu), \quad R(\lambda) = \lambda - i\hbar P,$$

exactly, to all orders in $\hbar$, and fails with $\lambda + i\hbar P$. Consequently the monodromy $T = L_N \cdots L_1$ satisfies RTT , and $[t(\lambda), t(\mu)] = 0$.

Proof. Write $L(\lambda) = \lambda E_{11} + L^0$ with $L^0 = L(0)$. In the leg notation the $(ij), (kl)$ entry of $L_a(\lambda)L_b(\mu)$ is $L_{ik}(\lambda)L_{jl}(\mu)$, and of $L_b(\mu)L_a(\lambda)$ it is the same product in the other order. So their difference is the matrix of commutators $-[L_{ik}, L_{jl}]$, which do not depend on λ, μ . The only non-zero ones are $[p, e^{\mp q}] = \pm i\hbar e^{\mp q}$, exactly $i\hbar$ times the Poisson brackets used in the proof of Proposition 9.9; that proof computed the whole matrix of brackets, so

$$L_b(\mu)L_a(\lambda) - L_a(\lambda)L_b(\mu) = -i\hbar (E_{11} \otimes L^0 - L^0 \otimes E_{11})P.$$

Next, $P L_b(\mu)L_a(\lambda) P = L_a(\mu)L_b(\lambda)$ (conjugating by P exchanges the legs), and expanding $L = \lambda E_{11} + L^0$ gives $L_a(\mu)L_b(\lambda) - L_a(\lambda)L_b(\mu) = (\mu - \lambda)(E_{11} \otimes L^0 - L^0 \otimes E_{11})$; the $L^0 \otimes L^0$ terms, in the same order on both sides, cancel. With $R = (\lambda - \mu) + si\hbar P$ the difference of the two sides of the relation is

$$(\lambda - \mu)(L_b L_a - L_a L_b) + si\hbar (P L_b L_a - L_a L_b P) = -(1 + s) i\hbar (\lambda - \mu)(E_{11} \otimes L^0 - L^0 \otimes E_{11})P,$$

using $PL_bL_a = L_a(\mu)L_b(\lambda)P$. It vanishes for $s = -1$ only. Sites commute, so [Theorem 14.2](#) and [Theorem 14.3](#) apply verbatim, now with $R(u)R(-u) = -(u^2 + \hbar^2)$. ■

The coefficients of $t(\lambda) = \sum_k c_k \lambda^{N-k}$ are the quantum integrals. They are the expressions of (9.4), now as operators, with no ordering ambiguity since each term has at most one factor per site: $c_1 = -\sum p_n$, and $H = \frac{1}{2}c_1^2 - c_2 = \frac{1}{2}\sum p_n^2 + \sum e^{q_n - q_{n+1}}$ is the quantum periodic Toda Hamiltonian, a function of the commuting c_k . Its spectral problem was reduced to one-variable quantisation conditions, for small N by Gutzwiller (1981) and for all N by Sklyanin's separation of variables (1985) · *cited*.

$\hbar \rightarrow 0$

To compare with [Lecture 9](#), divide by $\lambda - \mu$:

$$\frac{R(\lambda - \mu)}{\lambda - \mu} = \mathbb{1} + i\hbar r(\lambda - \mu), \quad r(\lambda) = -\frac{P}{\lambda},$$

exactly the r of [Proposition 9.9](#), and (14.1)'s expansion with no $O(\hbar^2)$ term. The limit needs a rule for turning operators into functions.

Lemma 14.5 (Symbols). Write an operator polynomial in the p 's with all p 's on the right, $F = \sum_\alpha f_\alpha(q) p^\alpha$, and call $\sigma(F) = \sum_\alpha f_\alpha(q) p^\alpha$, a function of (q, p) , its symbol. Then

$$\sigma(FG) = \sum_\alpha \frac{(-i\hbar)^{|\alpha|}}{\alpha!} \partial_p^\alpha \sigma(F) \partial_q^\alpha \sigma(G), \quad \text{so} \quad \sigma([F, G]) = i\hbar \{\sigma(F), \sigma(G)\} + O(\hbar^2).$$

Proof (★). Move each $p_n = -i\hbar \partial_{q_n}$ in F to the right through G 's functions of q by Leibniz's rule, $p_n^k g = \sum_j \binom{k}{j} (-i\hbar)^j (\partial_{q_n}^j g) p_n^{k-j}$. The first-order terms of $\sigma(FG) - \sigma(GF)$ are $-i\hbar \sum_n (\partial_{p_n} f \partial_{q_n} g - \partial_{p_n} g \partial_{q_n} f) = i\hbar \{f, g\}$ in the course's convention. ■

Corollary 14.6 (The classical limit of RTT). The symbols of the entries of the quantum monodromy are the entries of the classical monodromy (9.3). At order $\hbar$ the RTT relation of [Theorem 14.4](#) becomes

$$\{T_1(\lambda), T_2(\mu)\} = [r(\lambda - \mu), T_1(\lambda)T_2(\mu)], \quad r(\lambda) = -\frac{P}{\lambda} :$$

[Proposition 9.9](#) and [Theorem 9.7](#), sign for sign.

Proof. Each entry of T is a sum of products of entries of different site matrices, which commute, so its symbol is the classical product. Write RTT as $T_a T_b - T_b T_a = i\hbar(r T_b T_a - T_a T_b r)$ and take symbols. On the left the $(ij), (kl)$ entry is the commutator $[T_{ik}(\lambda), T_{jl}(\mu)]$, whose symbol is $i\hbar \{T_{ik}, T_{jl}\} + O(\hbar^2)$ by [Lemma 14.5](#): the entry of $i\hbar \{T_1, T_2\}$ in [Definition 9.2](#). On the right the symbols of $T_b T_a$ and $T_a T_b$ are both $T_1 T_2 + O(\hbar)$. Divide by $i\hbar$ and let $\hbar \rightarrow 0$. ■

Tracing over both legs, $\{t(\lambda), t(\mu)\} = 0$ is the order- $\hbar$ term of [Theorem 14.3](#). With $\lambda + i\hbar P$, false for this chain, the same steps would give $r = +P/\lambda$: the classical side decides the quantum sign. The check script repeats this computation for two and three sites.

14.7 What it buys

- *A commuting family for the Heisenberg chain*, from one relation for one site ([Theorems 14.1 to 14.3](#)). [Lecture 15](#) obtains the chain's Hamiltonian from its logarithmic derivative and diagonalises the family.
- *The Yang–Baxter equation, read a second time*: the Lax operator is Yang's R with a site as one of its spaces, and RLL is the equation of [Lecture 13 \(14.8\)](#).
- *G5, closed*: the quantum Toda chain's RTT relation becomes [Lecture 9's](#) quadratic bracket as $\hbar \rightarrow 0$, sign for sign ([Corollary 14.6](#)).

14.8 Caveats, in one place

- *Status*. [Theorems 14.1 to 14.4](#) and [Corollary 14.6](#) are proved; the proof of [Lemma 14.5](#) is a reading. The quantum Toda chain's spectral problem is cited.
- *What RTT does not give*. It gives commuting operators, not a Hamiltonian worth having or locality ([Lecture 15](#)), and not completeness, which stays [Lecture 10's](#) cited question. The independence of the quantum Toda integrals is not shown.
- *The classical limit is tested on Toda*; the Heisenberg chain has no small parameter.

Tutorial

14.9 Two sites by hand

For $N = 2$, $t(\lambda) = \text{tr}_a [(\lambda + iP_{a2})(\lambda + iP_{a1})] = 2\lambda^2 + 2i\lambda - \text{tr}_a (P_{a2}P_{a1})$, using $\text{tr}_a \mathbb{1} = 2$ and $\text{tr}_a P_{an} = 1$. By [\(14.3\)](#), $P_{a2}P_{a1} = P_{21}P_{a2}$, so $\text{tr}_a (P_{a2}P_{a1}) = P_{12}$ and

$$t(\lambda) = 2\lambda^2 + 2i\lambda - P_{12},$$

which commutes with itself at any two values, as it must: it involves only P_{12} . On the triplet $P_{12} = 1$, on the singlet -1 . Then $t(0) = -P_{12}$, the swap of the two sites, and $t(0)^{-1}t'(0) = (-P_{12})(2i) = -2iP_{12}$: [Lecture 15](#) finds the chain's Hamiltonian in this logarithmic derivative. Repeat for $N = 3$ and check that $t(0) = -iP_{32}P_{31}$.

Reading

14.10 The classical Yang–Baxter equation

The linear Sklyanin bracket [\(9.11\)](#) of [Lecture 9's](#) readings is an assumption about a Poisson bracket, so it must be consistent with the Jacobi identity. Working out what that requires produces the classical form of [Lecture 13's](#) equation.

Put three legs in play, $V \otimes V \otimes V$, write L_i for L in leg i at spectral parameter λ_i , and abbreviate $r_{ij} = r_{ij}(\lambda_i - \lambda_j)$. Two facts do the work: $[r_{ij}, L_k] = 0$ when $k \notin \{i, j\}$, because they act on disjoint legs; and $r_{21} = -r_{12}$, which [Remark 9.13](#) showed to be *sufficient* for antisymmetry of the Poisson bracket but which the bracket does not by itself force. It is therefore a hypothesis of [Theorem 14.7](#), not a consequence of [Definition 9.12](#), and is assumed from here on.

Theorem 14.7 (Jacobi and the CYBE). For a Lax matrix satisfying (9.11) with an r obeying the skew condition (9.12), the Jacobi identity for L_1, L_2, L_3 is equivalent to

$$[[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}], L_1 + L_2 + L_3] = 0. \quad (14.11)$$

In particular it holds whenever r satisfies the classical Yang–Baxter equation

$$[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] = 0. \quad (14.12)$$

Proof. Since r is numerical it passes through a Poisson bracket, so

$$\{\{L_1, L_2\}, L_3\} = [r_{12}, \{L_1, L_3\} + \{L_2, L_3\}] = [r_{12}, [r_{13}, L_1 + L_3] + [r_{23}, L_2 + L_3]].$$

Use $[r_{12}, L_3] = 0$ and the *matrix* Jacobi identity in the form $[A, [B, C]] = [[A, B], C] + [B, [A, C]]$ on each of the two terms. The first gives

$$[r_{12}, [r_{13}, L_1 + L_3]] = [[r_{12}, r_{13}], L_1 + L_3] + [r_{13}, [r_{12}, L_1]],$$

and the second the same with (13) $\rightarrow$ (23), $L_1 \rightarrow L_2$. Now add the two cyclic permutations of this, replacing r_{21}, r_{31}, r_{32} by $-r_{12}, -r_{13}, -r_{23}$ as (9.12) allows. The six leftover double commutators pair up, again by matrix Jacobi: for instance

$$[r_{13}, [r_{12}, L_1]] - [r_{12}, [r_{13}, L_1]] = -[[r_{12}, r_{13}], L_1].$$

Now collect. The first group contributes each double commutator of r 's twice, once from each of the two cyclic terms that produce it, and the two contributions carry different pairs of L_i :

$$[r_{12}, r_{13}] : L_1 + L_2 \text{ and } L_1 + L_3; \quad [r_{12}, r_{23}] : L_1 + L_2 \text{ and } L_2 + L_3; \quad [r_{13}, r_{23}] : L_1 + L_3 \text{ and } L_2 + L_3,$$

so their totals are $2L_1 + L_2 + L_3$, $L_1 + 2L_2 + L_3$ and $L_1 + L_2 + 2L_3$ respectively. The three surviving pairs of double commutators, computed as displayed above, contribute exactly $-[[r_{12}, r_{13}], L_1]$, $-[[r_{12}, r_{23}], L_2]$ and $-[[r_{13}, r_{23}], L_3]$, which remove the one doubled term in each line. All three are left with $L_1 + L_2 + L_3$, and that is (14.11). ■

Remark 14.8 (Machine-checked). The collection above is exactly the kind of bookkeeping in which a sign is lost. It is verified in `scripts/check-l8-algebra.py`, for $d = 2$, with three explicit legs and *symbolic* entries throughout — sixty free parameters, so the result is an identity and not a sample. The Jacobi sum equals $[X, L_1 + L_2 + L_3]$, and — the point of Remark 14.9 — is *not* identically zero on its own.

Remark 14.9 (Jacobi does not force the CYBE, and saying so matters). Theorem 14.7 gives (14.11), and (14.12) is *sufficient* for it, not implied by it: the Jacobi identity only requires the triple sum to *commute with* $L_1 + L_2 + L_3$, and for a particular L that is a weaker demand. What makes (14.12) the right thing to impose is that it mentions no L at all. One solution r then serves every model whose Lax matrix it fits, which is why the classification of r -matrices is a subject and the classification of integrable models is not.

One variant deserves its name in passing. Replacing the zero on the right of (14.12) by a specific r -independent term gives the *modified classical Yang–Baxter equation*, whose solutions generate the same structures over a slightly larger class of Lie algebras. Its theory belongs to Lie-bialgebra and Poisson–Lie theory, which this course does not enter; it is named here so that meeting the phrase elsewhere is not a surprise.

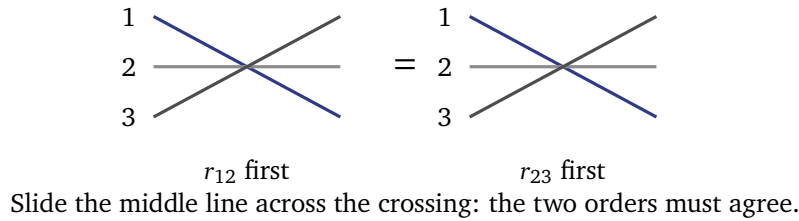


Figure 14.8: The classical Yang–Baxter equation. Three lines cross pairwise, and the two ways of resolving the triple crossing — sliding line 2 past the crossing of 1 and 3 either before or after — give the same answer. This is the classical form of Figure 13.1.

Reading

The rational solution

Proposition 14.10 (The rational r -matrix). $r(\lambda) = cP/\lambda$ satisfies the CYBE (14.12) for every constant c .

Proof. Write $\lambda_{ij} = \lambda_i - \lambda_j$. The permutation operators satisfy $P_{12}P_{13} = P_{13}P_{23} = P_{23}P_{12}$ and $P_{13}P_{12} = P_{12}P_{23} = P_{23}P_{13}$ — both are the two ways of cycling three legs. Calling the difference of those two products K , all three commutators are $\pm K$:

$$[P_{12}, P_{13}] = K, \quad [P_{12}, P_{23}] = -K, \quad [P_{13}, P_{23}] = K.$$

So the left of (14.12) is

$$c^2 K \left(\frac{1}{\lambda_{12}\lambda_{13}} - \frac{1}{\lambda_{12}\lambda_{23}} + \frac{1}{\lambda_{13}\lambda_{23}} \right) = c^2 K \frac{\lambda_{23} - \lambda_{13} + \lambda_{12}}{\lambda_{12}\lambda_{13}\lambda_{23}} = 0,$$

because $\lambda_{12} + \lambda_{23} = \lambda_{13}$ identically. ■

The cancellation is a statement about the three spectral parameters and nothing else, which is why c is free: the normalisation of r is fixed by the Lax matrix it is paired with, not by the CYBE. [Lecture 9](#) fixes it for the periodic Toda chain, $c = -1$ in (9.10); the open chain of [Lecture 9](#)'s readings needs a different, constant r .

Two further families solve (14.12): the *trigonometric* r -matrices, in which $1/\lambda$ is replaced by $\cot \lambda$ or $\coth \lambda$ and the permutation is deformed, and the *elliptic* ones, built from Weierstrass functions. Belavin and Drinfeld classified the solutions *under hypotheses this course has not set up*: for r valued in $\mathfrak{g} \otimes \mathfrak{g}$ with $\mathfrak{g}$ a finite-dimensional *simple* Lie algebra, in difference form, and non-degenerate in their technical sense (the residue at the simple pole is proportional to the Casimir of an invariant form), every solution is equivalent to a rational, trigonometric or elliptic one *· cited*, and the three degenerate into one another in limits. Those hypotheses are doing real work — dropped, the statement is false, and already fails for $d = 1$, where every commutator in (14.12) vanishes and any function of λ with a simple pole at the origin solves it. The same trichotomy reappears in the Calogero–Moser potentials (optional lecture B) and in the quantum chains XXX, XXZ and XYZ — it is the same list every time, which is not a coincidence.

Reading

The open chain's r is not a solution

The open chain's constant r of (9.22) (Lecture 9's readings) is skew, $PrP = -r$, which is (9.12) for a constant r ; so Theorem 14.7 applies to it. Does it solve the classical Yang–Baxter equation (14.12)? **No.** Its left-hand side is

$$[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] = -\frac{1}{4} [P_{12}, P_{13}] \neq 0. \quad (14.13)$$

This identity is *cited*, not proved here: Babelon–Bernard–Talon, eq. (4.32), state it in operator form, and `scripts/check-l9-algebra.py` checks it symbolically for $N = 2, 3$. An equation of this shape, with a fixed non-zero right-hand side in place of the zero, is the *modified* CYBE named above.

And yet the Jacobi identity holds, as it must: the bracket is the canonical one on $\mathbb{R}^{2N}$. Theorem 14.7 asks only that the left of (14.13) commute with $L_1 + L_2 + L_3$, and it does, for *every* matrix L : P_{12} commutes with $L_1 + L_2$ (it only swaps them) and with L_3 (another leg), so it commutes with the sum, and so does P_{13} , and so does their commutator. This is Remark 14.9 in a real model: the CYBE is sufficient for Jacobi, and the Toda chain satisfies Jacobi without it.

How unique is r ? Adding $x\mathbb{1} + yP$ changes nothing in (9.21), since both commute with $L_1 + L_2$ (P only swaps the two legs). Babelon–Bernard–Talon, §4.7, derive (9.22) rather than check it, and show it is the only skew constant solution; `scripts/check-rmatrix.py` (V8) finds, at sampled points for $N = 3$, no constant solutions beyond $r + x\mathbb{1} + yP$. Of these, the skew one is r itself, which fails the CYBE, while $r \pm \frac{1}{2}P$ *do* solve the CYBE (Babelon–Bernard–Talon, (4.33); checked for $N = 2, 3$) and are not skew, so Theorem 14.7 as proved does not cover them. So no constant r for the open chain is both skew and a solution of the CYBE: the route “skew plus CYBE, hence Jacobi” is closed here, and Jacobi holds by the weaker condition above. That is not a defect of the model; it is why Lie-bialgebra theory has a modified equation.

Reading

14.11 Reordering three monodromies

Three auxiliary spaces 1, 2, 3 at parameters $\lambda_1, \lambda_2, \lambda_3$ give three relations (14.6), one for each pair, with $R_{ij} = R_{ij}(\lambda_i - \lambda_j)$.

Proposition 14.11 (Reordering three monodromies). *Let T satisfy (14.6) with an invertible R . Reversing $T_1T_2T_3$ one neighbouring pair at a time gives, along the two possible routes,*

$$T_1T_2T_3 = W_A (T_3T_2T_1) W_A^{-1} = W_B (T_3T_2T_1) W_B^{-1}, \quad W_A = R_{12}R_{13}R_{23}, \quad W_B = R_{23}R_{13}R_{12}.$$

Hence $W_B^{-1}W_A$ commutes with every $T_3T_2T_1$.

Proof. Write (14.6) as $T_iT_j = R_{ij}T_jT_iR_{ij}^{-1}$ for $i < j$. Route A exchanges the pairs 12, then 13, then 23: $T_1T_2T_3 = R_{12}T_2T_1T_3R_{12}^{-1} = R_{12}R_{13}T_2T_3T_1R_{13}^{-1}R_{12}^{-1} = W_A T_3T_2T_1 W_A^{-1}$, each R_{ij} commuting with the monodromy it does not touch. Route B exchanges 23, 13, 12 and gives W_B in the same way. Equating the two results gives the statement. ■

For a model that satisfies *RTT* this is automatic — the two routes rewrite one operator. It matters when *RTT* is taken as the *definition* of an algebra generated by the entries of T , as a Sklyanin bracket can be posited ([Lecture 9](#)'s readings): then the two routes yield the commutation condition of the proposition among the generators, and $W_A = W_B$ makes it automatic. That is sufficient, not necessary: for $R = \sigma^x \otimes \sigma^z$, which fails the Yang–Baxter equation, $W_B = -W_A$ and the condition is again automatic (the check script). It is [Remark 14.9](#) quantised: there the Jacobi identity required a combination of r 's only to commute with $L_1 + L_2 + L_3$, here the associativity of the products requires $W_B^{-1}W_A$ only to commute with the triple products, and for a particular model that is a weaker demand. What makes the Yang–Baxter equation the right thing to impose is, again, that it mentions no model. (The recipe $L_{an}(\lambda) = -R_{an}(-\lambda)$ of (14.7) used $R_{12} = R_{21}$; for a solution without that symmetry it is $L_{an}(\lambda) = -R_{na}(-\lambda)$.) For the Heisenberg chain nothing is left to choose: the check script finds that (14.4) fixes R up to a scalar for one and two sites, at generic λ, μ .

Reading

14.12 The Yang–Baxter equation at order $\hbar^2$, and more

YBE $\rightarrow$ *CYBE*. Give each line i a parameter λ_i and write $R_{ij} = R_{ij}(\lambda_i - \lambda_j)$, so (14.8) is $R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$. Suppose $R = \mathbb{1} + i\hbar r + \hbar^2 s + O(\hbar^3)$, in difference form. The terms of order $\hbar$ are $r_{12} + r_{13} + r_{23}$ on both sides. Those of order $\hbar^2$ containing s are also the same on both sides; the rest are $(i\hbar)^2$ times the products of two r 's in the order of the factors, so the difference of the two sides at order $\hbar^2$ is

$$-\hbar^2 \left([r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] \right) :$$

a solution of the Yang–Baxter equation has a classical r that solves the CYBE (14.12), in the per-leg form of § 14.10. For $R = \lambda - i\hbar P$, normalised by $1/\lambda$ (which does not change (14.8), the scalars being the same three on both sides), $r = -P/\lambda$ is [Proposition 14.10](#) at $c = -1$. This is the picture [Figure 14.8](#) promised: the classical slide is the first correction to the quantum one · *proved; the check script repeats it with generic matrices.*

The quantum determinant. At $\lambda - \mu = i$, $R(i) = i(1 - P)$ is $2i$ times the projector onto the antisymmetric part of $\mathbb{C}^2 \otimes \mathbb{C}^2$, and *RTT* says that $T_a(\mu + i)T_b(\mu)$ maps the antisymmetric line to itself. The operator by which it multiplies that line (central; for XXX a number) is the *quantum determinant*,

$$\text{qdet } T(\mu) = A(\mu + i)D(\mu) - B(\mu + i)C(\mu),$$

the counterpart of $\det T = 1$ for Toda (9.3). It commutes with every entry of T (Korepin, Bogoliubov and Izergin, §VII.8, in their conventions). For the universal algebra defined by *RTT* on a normalised formal series $T(u) = \mathbb{1} + T^{(1)}u^{-1} + \dots$, the Yangian $Y(\mathfrak{gl}_2)$, the coefficients of the quantum determinant generate the centre (Molev, ch. 1) · *cited*; a spin chain gives a representation of it. For the Heisenberg chain it is the number $[\mu(\mu + 2i)]^N$ (the check script, $N \leq 3$). [Lecture 15](#) meets its two factors as the eigenvalues of A and D on the reference state.

Other spins. For a spin s at each site, $L_{an}(\lambda) = \lambda + \frac{i}{2} + i\sigma_a \cdot \mathbf{S}_n$ satisfies (14.4) with the same R , by a verification that uses only the commutation relations of $\mathbf{S}$ · *cited: Korepin, Bogoliubov and Izergin, §VIII.4, (4.6)–(4.8), in their conventions; the check script verifies $s = 1$.* The transfer matrix still commutes with itself; but the regular point is lost, since $L(0)$ is

no longer a permutation, and a local Hamiltonian needs a transfer matrix with a spin- s auxiliary space (fusion: Kulish, Reshetikhin and Sklyanin 1981; not treated here).

Reading

14.13 Why the zero-curvature equation is not what is quantised

Lecture 8's and Lecture 9's readings derived the conservation of $t(\lambda)$ from the zero-curvature equation (8.23), $\dot{L}_n = V_{n+1}L_n - L_nV_n$, and Lecture 9 proved, separately, the Sklyanin bracket. This lecture quantised only the second. The zero-curvature equation is the equation of motion under one Hamiltonian, and in quantum mechanics equations of motion are not postulated: they follow from $[\cdot, H]$ once H is known. What has to be quantised is the *kinematics*, the brackets, and from them the commuting family and its Hamiltonians follow. The site-by-site transport survives as the monodromy (14.5), and the bracket as RTT . The Heisenberg equation for $T(\lambda)$ under H is then a consequence, and $[H, t(\lambda)] = 0$ is Theorem 14.3, not an input. That is the reason the classical limit of § 14.6 was taken in the bracket, not in the equation of motion.

Reading

14.14 The Toda pair, quantised

Lecture 12 set the Toda displacement beside Lecture 11's θ' as a counterpart. For the Toda pair itself the link can be checked. In Proposition 12.1 the relative coordinate $x = q_1 - q_2$ and its momentum $p = \frac{1}{2}(p_1 - p_2)$ have $H_{\text{rel}} = p^2 + e^x$, and the relative speed is $v = 2p$. Quantised,

$$-\hbar^2\psi'' + e^x\psi = p^2\psi.$$

With $\hbar = 1$ and $k = p$, the solution that decays under the wall is $\psi = K_{2ik}(2e^{x/2})$, the modified Bessel function: the chain rule turns this equation into Bessel's. Far from the wall, $x \rightarrow -\infty$, the small-argument form of K_ν gives · cited: DLMF §§10.25–10.27

$$\psi \approx \frac{1}{2} [\Gamma(-2ik) e^{ikx} + \Gamma(2ik) e^{-ikx}] :$$

an incoming wave e^{ikx} , reflected with the phase $\Gamma(2ik)/\Gamma(-2ik)$. Since $\Gamma(-2ik) = \overline{\Gamma(2ik)}$, the stationary-phase argument of § 11.3 puts the incoming packet on $x = 2kt + x_w$ and the outgoing one on $x = -2kt + x_w$, with $x_w = 2 \operatorname{Re} \psi_0(2ik)$ and $\psi_0 = \Gamma'/\Gamma$ the digamma function: the packet moves as if reflected by a hard wall at x_w . In the language of § 11.3, with $k = p_1 - p_2 = 2p$ the momentum difference, the reflected wave's phase is $\theta(k) = 2 \arg \Gamma(ik)$ up to a constant, and $\theta'(k) = 2 \operatorname{Re} \psi_0(ik) = x_w$: the wall is θ' . Restoring $\hbar$ shifts x by $2 \ln \hbar$ and replaces k by $p/\hbar$, so

$$x_w = 2 \ln \hbar + 2 \operatorname{Re} \psi_0(2ip/\hbar) \longrightarrow 2 \ln(2p) = 2 \ln v \quad (\hbar \rightarrow 0),$$

by Stirling's formula, $\operatorname{Re} \psi_0(iy) = \ln y + O(y^{-2})$. The classical orbit of Proposition 12.1, $x = \ln v^2 - v|t| + o(1)$, has its effective wall exactly there, and that wall is what displaces each momentum line by $\mp 2 \ln v$. So the classical displacement is the $\hbar \rightarrow 0$ limit of the quantum one (scripts/check-116-algebra.py checks the Bessel equation, the asymptotic

expansion and the effective-wall limit; the stationary-phase step is argued, not checked; Sutherland §4.2 does the same for the whole hyperbolic family) · *sketched*.

Summary.

- The Lax operator $L_{an}(\lambda) = \lambda + iP_{an} = -R_{an}(-\lambda)$ satisfies *RLL* with $R(\lambda) = \lambda - iP$ (Theorem 14.1), which is Lecture 13's Yang–Baxter equation.
- The train (Theorem 14.2): the monodromy satisfies *RTT*, and $[t(\lambda), t(\mu)] = 0$ (Theorem 14.3).
- The classical limit (Theorem 14.4 and Corollary 14.6): the quantum Toda chain satisfies *RLL* exactly with $R = \lambda - i\hbar P$, and at order $\hbar$ its *RTT* relation is Lecture 9's quadratic bracket with $r = -P/\lambda$.
- Next: Lecture 15 finds the Heisenberg Hamiltonian in $t(\lambda)$ and diagonalises the family by the algebraic Bethe ansatz.

Tutorial. Exercise Sheet 14.

References

E. K. Sklyanin, L. A. Takhtajan and L. D. Faddeev, *Theor. Math. Phys.* 40 (1979) 688 (the method); L. D. Faddeev, *How algebraic Bethe ansatz works for integrable model*, Les Houches lectures 1995, arXiv:hep-th/9605187, §§1–3; V. E. Korepin, N. M. Bogoliubov and A. G. Izergin, *Quantum Inverse Scattering Method and Correlation Functions*, Cambridge University Press (1993), §§VI.1–VI.5, §VII.8 (the quantum determinant), §VIII.4 (higher spin); M. C. Gutzwiller, *Ann. Phys.* 133 (1981) 304 and E. K. Sklyanin, *Lect. Notes Phys.* 226 (1985) 196 (the quantum Toda chain); A. Molev, *Yangians and Classical Lie Algebras*, American Mathematical Society (2007), Chapter 1 (the Yangian and its centre); O. Babelon, D. Bernard and M. Talon, *Introduction to Classical Integrable Systems*, Cambridge University Press (2003), Ch. 2–3 (the classical Yang–Baxter equation) and §4.7 (the open chain's r). **The conventions for the ordering of *RTT*, of the monodromy and for the sign in R differ between these sources; see § 14.2.**